

# Flow and Rate Control

Window Flow Control · Token Bucket · Rate Control · Max-Min ·  
Proportional Fair

2026-06-08

|          |                                           |           |
|----------|-------------------------------------------|-----------|
| <b>1</b> | :                                         | <b>2</b>  |
| <b>2</b> | <b>Window Flow Control</b>                | <b>2</b>  |
| 2.1      | .....                                     | 2         |
| 2.2      | .....                                     | 4         |
| 2.3      | .....                                     | 4         |
| <b>3</b> | <b>Rate Control — Token(Leaky) Bucket</b> | <b>5</b>  |
| 3.1      | .....                                     | 5         |
| 3.2      | ( ) .....                                 | 5         |
| 3.3      | ( ) .....                                 | 5         |
| 3.4      | (balance $\rightarrow$ ) .....            | 6         |
| 3.5      | $p_0$ : (conservation) .....              | 7         |
| 3.6      | permit .....                              | 7         |
| 3.7      | $W$ trade-off .....                       | 8         |
| 3.8      | ( ) .....                                 | 8         |
| <b>4</b> | <b>Rate Control</b>                       | <b>10</b> |
| 4.1      | .....                                     | 10        |
| 4.2      | Overflow .....                            | 10        |
| 4.3      | (KKT ) .....                              | 10        |
| 4.4      | — .....                                   | 11        |
| 4.5      | 3- - ( ) .....                            | 11        |
| <b>5</b> | <b>Max-Min Fairness</b>                   | <b>13</b> |
| 5.1      | .....                                     | 13        |
| 5.2      | Bottleneck .....                          | 14        |

|          |                                     |           |
|----------|-------------------------------------|-----------|
| 5.3      | Water-filling                       | 14        |
| 5.4      |                                     | 14        |
| <b>6</b> | <b>Proportional Fair Scheduling</b> | <b>15</b> |
| 6.1      |                                     | 15        |
| 6.2      | proportional fair                   | 15        |
| 6.3      | 3- + Multiuser diversity            | 16        |
| <b>7</b> |                                     | <b>16</b> |

# 1 :

“ , ” .

|                             |                  |                                 |
|-----------------------------|------------------|---------------------------------|
| Window flow control         | (outstanding)    | round-trip                      |
|                             | $W$              |                                 |
| Rate control (token bucket) | permit $r$ , $W$ | Markov chain .                  |
| / rate control              | $r_w$            | , max-min / proportional fair . |

, ( · ) .

!

, . [ ] “ ” .

## 2 Window Flow Control

### 2.1

End-to-end window ACK  $W$  .

- $d$ : round-trip delay (sec)
- $X$ : (sec) =
- $W$ : window size ( )

window  $d$ ,  $W$  window  $W/d$ ,  $1/X$ . :

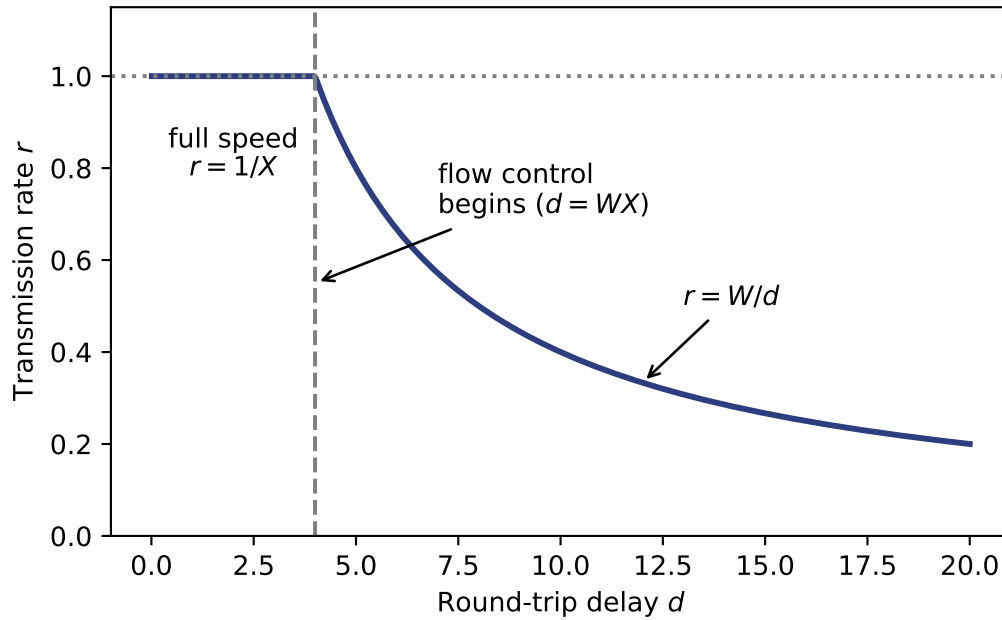
$$r = \min\left[\frac{1}{X}, \frac{W}{d}\right] \quad [\text{packets/sec}]$$

- $d \leq WX$  :  $r = 1/X$  (full speed, window )
- $d > WX$  :  $r = W/d$  ( window throttle = flow control )

```
import numpy as np
import matplotlib.pyplot as plt

X, W = 1.0, 4.0
d = np.linspace(0.01, 20, 500)
r = np.minimum(1/X, W/d)

fig, ax = plt.subplots()
ax.plot(d, r, lw=2.2, color="#2c3e80")
ax.axvline(W*X, ls="--", color="gray")
ax.axhline(1/X, ls=":", color="gray")
ax.annotate("full speed\n$r=1/X$", xy=(W*X/2, 1/X),
            xytext=(W*X/2, 0.78), ha="center")
ax.annotate("$r=W/d$", xy=(12, W/12), xytext=(13, 0.5),
            arrowprops=dict(arrowstyle="->"))
ax.annotate("flow control\nbegins ($d=WX$)", xy=(W*X, 0.55),
            xytext=(7, 0.7), arrowprops=dict(arrowstyle="->"))
ax.set_xlabel("Round-trip delay $d$"); ax.set_ylabel("Transmission rate $r$")
ax.set_ylim(0, 1.15); plt.tight_layout()
```



1: RTT( $d$ ) .  $d = WX$  full-speed flow control .

## 2.2

1. .  $d$   $W$   $r = W/d$   $\rightarrow$  window RTT ( ).
2. . full speed  $W \geq d/X = d \cdot (\text{link rate})$ , **bandwidth-delay product** in-flight .

## 2.3

- **Node-by-node (hop-by-hop):** window  $\rightarrow$  **backpressure** , end-to-end .
- **Isarithmic:** permit  $\rightarrow$  .

### 💡 [ ] Window

- :  $r = \min[1/X, W/d]$ .
- :  $d, X, W$   $r$  , full-speed  $d \leq WX$  , .
- “ ”, “BDP ” .

### 3 Rate Control — Token(Leaky) Bucket

.  $\rightarrow$   $\rightarrow$  DTMC  $\rightarrow$   $\rightarrow$  .

#### 3.1

- **permit(token)** .
- Token  $1/r$  **1** (  $W$  ) . token .
- token .

#### 3.2 ( )

token embedding Markov chain.  $i$ :

- $i = 0, 1, \dots, W$  : , token  $(W - i)$  . ( $i = 0 =$  ,  $i = W =$  token 0 )
- $i = W + 1, W + 2, \dots$  : token 0 ,  $(i - W)$  .

$\lambda$  Poisson slot( $1/r$ )  $k$

$$a_k = \frac{(\lambda/r)^k}{k!} e^{-\lambda/r}, \quad k = 0, 1, 2, \dots$$

$\lambda/r = \rho$  Poisson.

#### 3.3 ( )

slot token  $-1,$   $+A.$   $j \geq 1$  token  $i = j - 1 + A,$   $A = i - j + 1$ :

$$P_{ji} = a_{i-j+1} \quad (j \leq i + 1), \quad 0 \text{ otherwise.}$$

$0( )$  :  $A = 0(\text{token } ) \cdot A = 1( )$   $0$  :

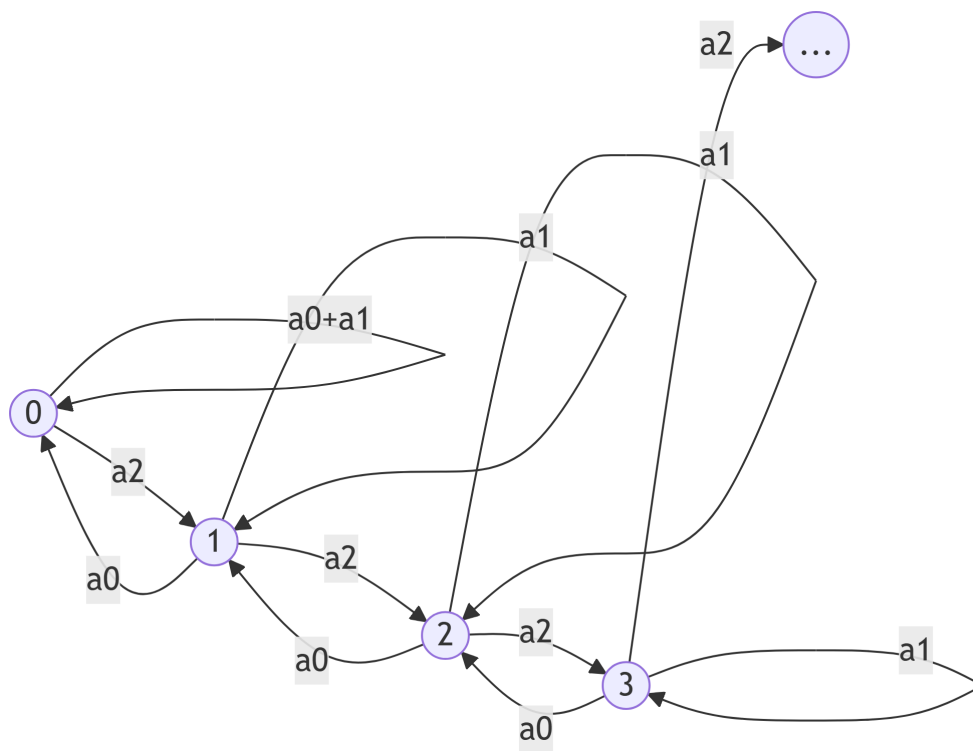
$$P_{0i} = \begin{cases} a_0 + a_1, & i = 0 \\ a_{i+1}, & i \geq 1 \end{cases}$$

$a_0 + a_1$   $W$  . M/G/1 chain *skip-free to the left* .

```

flowchart LR
    S0((0)) -->|a0+a1| S0
    S0 -->|a2| S1((1))
    S1 -->|a0| S0
    S1 -->|a1| S1
    S1 -->|a2| S2((2))
    S2 -->|a0| S1
    S2 -->|a1| S2
    S2 -->|a2| S3((3))
    S3 -->|a0| S2
    S3 -->|a1| S3
    S3 -->|a2| S4(("..."))

```



2: Token bucket DTMC ( ).  $a_0, / a_1, a_2, \dots$

### 3.4 (balance $\rightarrow$ )

$$j = 0 \quad a_{i+1} = a_{i-0+1} \quad ,$$

$$p_0 = a_0 p_1 + (a_0 + a_1) p_0, \quad p_i = \sum_{j=0}^{i+1} a_{i-j+1} p_j \quad (i \geq 1).$$

0

$$p_0(1 - a_0 - a_1) = a_0 p_1 \Rightarrow p_1 = \frac{1 - a_0 - a_1}{a_0} p_0,$$

$$p_2, p_3, \dots \quad p_0 \quad .$$

### 3.5 $p_0$ : (conservation)

. Token  $r$  ,  $0$   $(a_0)$   $= r(1 - p_0 a_0)$ . throughput  
 $= \lambda :$

$$r(1 - p_0 a_0) = \lambda \Rightarrow p_0 = \frac{r - \lambda}{r a_0}$$

**i**

$$p_0 > 0 \Leftrightarrow \lambda < r. \quad \text{token} \quad (\rho = \lambda/r < 1).$$

### 3.6 permit

$$= \max(0, i - W), \quad N_w = \sum_{j>W} p_j(j - W). \quad \text{Little} \quad N_w = \lambda T :$$

$$T = \frac{1}{\lambda} \sum_{j=W+1}^{\infty} p_j(j - W)$$

**!**

r

Little  $\lambda$   $(r)$ .  $\lambda \rightarrow r$  , “average packet delay”  $N_w/\lambda$  .

### 3.7 $W$ trade-off

- $W \rightarrow$  permit  $(\quad, \quad)$
- $W \rightarrow (\quad)$

$r$ ,  $W$ .

### 3.8 $(\quad)$

:  $W = 0$ (token  $\quad$ ), slot  $2$ .

$$a_0 = 0.4, \quad a_1 = 0.4, \quad a_2 = 0.2, \quad \rho = \mathbb{E}[A] = 0.4 + 2(0.2) = 0.8 < 1.$$

$W = 0$   $n =$ .

Balance  $(\quad)$

$$\begin{aligned} p_0 &= (a_0 + a_1)p_0 + a_0p_1 = 0.8p_0 + 0.4p_1 & \Rightarrow p_1 &= 0.5p_0 \\ p_1 &= a_2p_0 + a_1p_1 + a_0p_2 = 0.2p_0 + 0.4p_1 + 0.4p_2 & \Rightarrow p_2 &= 0.25p_0 \\ p_2 &= a_2p_1 + a_1p_2 + a_0p_3 & \Rightarrow p_3 &= 0.125p_0 \end{aligned}$$

$$0.5 : p_n = p_0(0.5)^n. \quad \sum p_n = 2p_0 = 1 \rightarrow p_0 = 0.5,$$

$$p_n = (0.5)^{n+1}.$$

$$: p_0 = \frac{r - \lambda}{ra_0} = \frac{1 - \rho}{a_0} = \frac{0.2}{0.4} = 0.5.$$

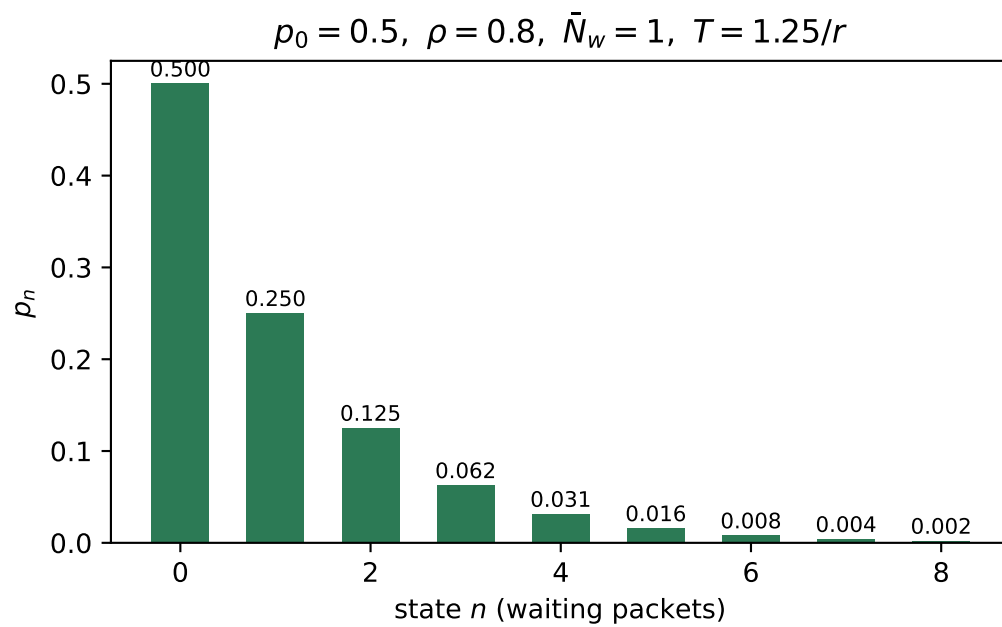
.  $W = 0$   $N_w = \mathbb{E}[n] = \sum_n n (0.5)^{n+1} = 1$ . Little :

$$T = \frac{N_w}{\lambda} = \frac{1}{0.8r} = \frac{1.25}{r} \quad (= 1.25 \text{ slot}).$$

```
import numpy as np, matplotlib.pyplot as plt
n = np.arange(0, 9)
p = 0.5**(n+1)
fig, ax = plt.subplots()
ax.bar(n, p, color="#2c7a55", width=0.6)
for xi, pi in zip(n, p):
    ax.text(xi, pi+0.008, f"{pi:.3f}", ha="center", fontsize=8)
ax.set_xlabel("state $n$ (waiting packets)"); ax.set_ylabel("$p_n$")
```



```
ax.set_title(r"$p_0=0.5, \rho=0.8, \bar{N}_w=1, T=1.25/r$")
plt.tight_layout()
```



3:  $p_n = (0.5)^{n+1} ( )$ .

💡 [ ] Token bucket —

1. (token ).
2.  $P_{ji} = a_{i-j+1}, P_{00} = a_0 + a_1$  MC .
3. balance / .
4.  $p_0 = \frac{r - \lambda}{ra_0}$  .
5.  $N_w/\lambda$  (Little).

“ (token vs )” — slot . (a) · (c) .

## 4 Rate Control

### 4.1

class  $w$        $\bar{r}_w$ ,       $r_w \leq \bar{r}_w$ :

$$\min \sum_{(i,j)} D_{ij}(F_{ij}) + \sum_w e_w(r_w) \quad \text{s.t.} \quad \sum_{p \in P_w} x_p = r_w, \quad x_p \geq 0, \quad 0 \leq r_w \leq \bar{r}_w.$$

- $\sum D_{ij}(F_{ij})$ : (routing).
- $\sum e_w(r_w)$ : ( $r_w$  → “ ”).

↑,      ↑.

### 4.2 Overflow

$$y_w = \bar{r}_w - r_w \geq 0 \quad \rightarrow \quad :$$

$$\min \sum D_{ij}(F_{ij}) + \sum_w E_w(y_w), \quad \sum_p x_p + y_w = \bar{r}_w, \quad x_p, y_w \geq 0,$$

$$E_w(y_w) = e_w(\bar{r}_w - y_w), \quad e'_w(r_w) = -(a_w/r_w)^{b_w}.$$

### 4.3 (KKT )

$$d_p^* = \sum_{(i,j) \in P} D'_{ij}(F_{ij}^*) \quad ,$$

$$\begin{aligned} x_p^* > 0 &\Rightarrow d_p^* \leq d_{p'}^* \quad (\forall p' \in P_w) \text{ and } d_p^* \leq -e'_w(r_w^*), \\ r_w^* < \bar{r}_w &\Rightarrow -e'_w(r_w^*) \leq d_p^* \quad (\forall p \in P_w). \end{aligned}$$

:      ( )      (= routing ),       $-e'_w$ .

#### 4.4 —

$$C, \quad \bar{r}, \quad r, y = \bar{r} - r. \quad M/M/1 \quad D(r) = \frac{r}{C-r}, \quad \frac{a}{r} :$$

$$J(r) = \frac{r}{C-r} + \frac{a}{r}, \quad J'(r) = \frac{C}{(C-r)^2} - \frac{a}{r^2}.$$

$$J' = 0 :$$

$$\frac{\sqrt{C}}{C-r} = \frac{\sqrt{a}}{r} \Rightarrow r(\sqrt{C} + \sqrt{a}) = C\sqrt{a} \Rightarrow \boxed{r^* = \frac{C\sqrt{a}}{\sqrt{a} + \sqrt{C}}}$$

$$\text{flow control} \quad (r = \bar{r} \quad):$$

$$\frac{C}{(C-\bar{r})^2} < \frac{a}{\bar{r}^2} \iff \bar{r} < \frac{C\sqrt{a}}{\sqrt{a} + \sqrt{C}} = r^*.$$



$$r^* = \frac{C\sqrt{a}}{\sqrt{a} + \sqrt{C}} \cdot \frac{a}{a+C} = \frac{\sqrt{a}}{\sqrt{a} + \sqrt{C}}.$$

#### 4.5 3- - ( )

$$\text{OD, } C_1, C_2, C_3 \quad \text{one-hop}, \quad r :$$

$$\min \sum_{i=1}^3 \frac{f_i}{C_i - f_i} \quad \text{s.t.} \quad \sum_i f_i = r, \quad f_i \geq 0.$$

$$\text{Lagrangian } \mathcal{L} = \sum \frac{f_i}{C_i - f_i} - \nu(\sum f_i - r),$$

$$\frac{C_i}{(C_i - f_i)^2} = \nu \Rightarrow C_i - f_i = \sqrt{C_i/\nu} \Rightarrow f_i = C_i - \sqrt{\frac{C_i}{\nu}}.$$

$$\sum f_i = r \quad :$$

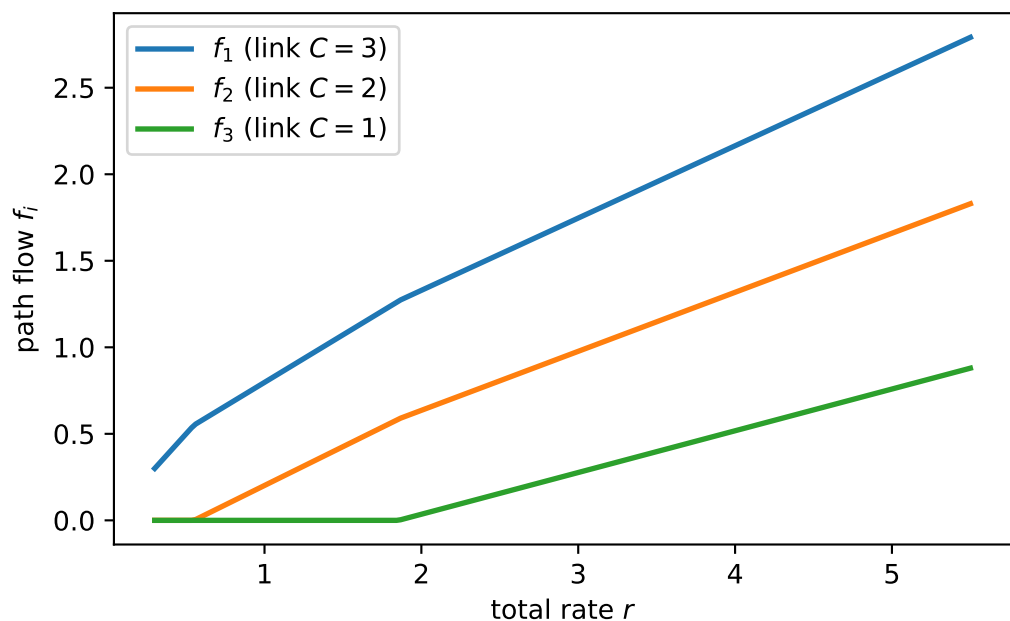
$$\sqrt{\nu} = \frac{\sum_i \sqrt{C_i}}{\sum_i C_i - r}, \quad \boxed{f_i = C_i - \sqrt{C_i} \frac{\sum_j C_j - r}{\sum_j \sqrt{C_j}}}$$

(  $f_i < 0$                        $f_i = 0$                       .)

```
import numpy as np, matplotlib.pyplot as plt

C = np.array([3.0, 2.0, 1.0])
def opt_flows(r):
    active = np.array([True, True, True])
    for _ in range(3):
        Ca = C[active]
        s = np.sum(np.sqrt(Ca)) / (np.sum(Ca) - r)
        f = np.zeros(3)
        f[active] = Ca - np.sqrt(Ca)/s
        if np.all(f[active] >= -1e-9):
            f[f < 0] = 0; return f
        active = f > 1e-9
    return np.clip(f, 0, None)

R = np.linspace(0.3, 5.5, 200)
F = np.array([opt_flows(r) for r in R])
fig, ax = plt.subplots()
for i, c in enumerate(C):
    ax.plot(R, F[:, i], lw=2, label=f"$f_{i+1}$ (link $C={c:g}$)")
ax.set_xlabel("total rate $r$"); ax.set_ylabel("path flow $f_i$")
ax.legend(); plt.tight_layout()
```



4:  $r$  ,  $(C = [3, 2, 1])$

💡 [ ] rate/routing

- $\frac{C_i}{(C_i - f_i)^2} + \sum f_i = r$ .
- / ( ).
- flow  $\rightarrow$  , .

## 5 Max-Min Fairness

$$\cdot \cdot , r_p, F_a = \sum_{p \ni a} r_p \leq C_a.$$

### 5.1

$r$  feasible ,  $r_p$  —  $r_{p'} \leq r_p$  — **max-min fair**.

: lexicographic .

## 5.2 Bottleneck

$a \in P$  bottleneck:  $F_a = C_a$  ( )  $a \in P$  ( $r_p \geq r_{p'}$ ).

**Proposition:**  $r$  max-min fair  $\Leftrightarrow$  bottleneck .

## 5.3 Water-filling

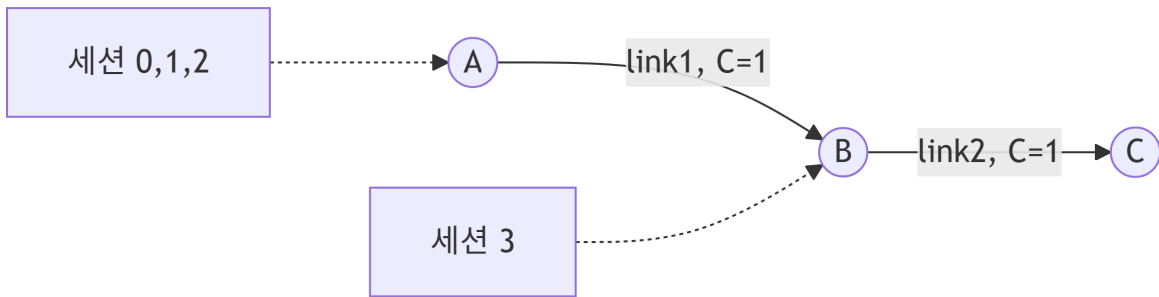
$k = 1, F_a^0 = 0, r_p^0 = 0, P^1 = P, A^1 = A.$

1.  $n_a^k = \frac{C_a - F_a^{k-1}}{r_p^k}$  .
2.  $\tilde{r}^k = \min_{a \in A^k} \frac{C_a - F_a^{k-1}}{n_a^k} \rightarrow$  bottleneck.
3.  $r_p^k = r_p^{k-1} + \tilde{r}^k$  (  $p \in P$  ), .
4.  $F_a^k = \sum_{p \ni a} r_p^k$ .
5.  $A^{k+1} = \{a : C_a - F_a^k > 0\}.$
6.  $P^{k+1} = \{p : r_p^k > 0\}.$
7.  $k \leftarrow k + 1.$  8.  $P$  .

## 5.4

flowchart LR

```
A((A)) -->|"link1, C=1"| B((B))
B -->|"link2, C=1"| D((C))
S012[" 0,1,2"] -.-> A
S3[" 3"] -.-> B
```



5: ( 1). 0,1,2 , 3 .

$0 \cdot 1 \cdot 2 \rightarrow 1/3$ .  $3 \quad 1 - 1/3 = 2/3 \rightarrow r_3 = 2/3$ . max-min ( ).

### 💡 [ ] Max-Min

- ( ) bottleneck (iff) .
- **water-filling** .
- “max-min” — .

## 6 Proportional Fair Scheduling

. throughput fairness (max-min ).

### 6.1

$\text{DRC}_k(t)$ ,  $R_k(t)$ . slot

$$k^* = \arg \max_k \frac{\text{DRC}_k(t)}{R_k(t)},$$

$$R_k(t+1) = \begin{cases} \left(1 - \frac{1}{t_c}\right) R_k(t) + \frac{1}{t_c} \text{DRC}_k(t), & k \\ \left(1 - \frac{1}{t_c}\right) R_k(t), & k \end{cases}$$

“ ” .

### 6.2 proportional fair

( ) :

$$P = \arg \max \sum_{k \in U} \log R_k.$$

$\partial(\log R_k)/\partial R_k = 1/R_k$  — 1 , . slot  $\text{DRC}_k/R_k \quad \sum \log R_k$   
(gradient).

:  $\sum_k \Delta R_k/R_k \leq 0$  ( ).

### 6.3 3- + Multiuser diversity

fading throughput, ( ). PF  $R_k$  peak  $\rightarrow$   
 throughput (multiuser diversity).

| Max-Min    | Proportional Fair   |
|------------|---------------------|
| ( )        | ( )                 |
| throughput | throughput-fairness |

💡 [ ] Proportional Fair

- $\sum \log R_k$   $\arg \max \text{DRC}_k / R_k$ , 3 “ ”.
- max-min 1 ( vs , ) .

## 7

! ( )  $\rightarrow$  ( )

|                   | ( )                    |                      |
|-------------------|------------------------|----------------------|
| Window            | $r = \min[1/X, W/d]$ . | / full-speed         |
| Token bucket      | chain + Little         | MC $p_0, T$          |
| rate/routing      | KKT                    | .                    |
| Max-Min           | bottleneck iff         | <b>water-filling</b> |
| Proportional Fair | $\sum \log R_k$        | + 3-                 |

token bucket( 3.8) ( 4.5) , . token bucket “  $\rightarrow$   
 MC  $\rightarrow$  balance  $\rightarrow N_w/\lambda$ ” 4 , “ +  $\sum f_i = r$ ” .